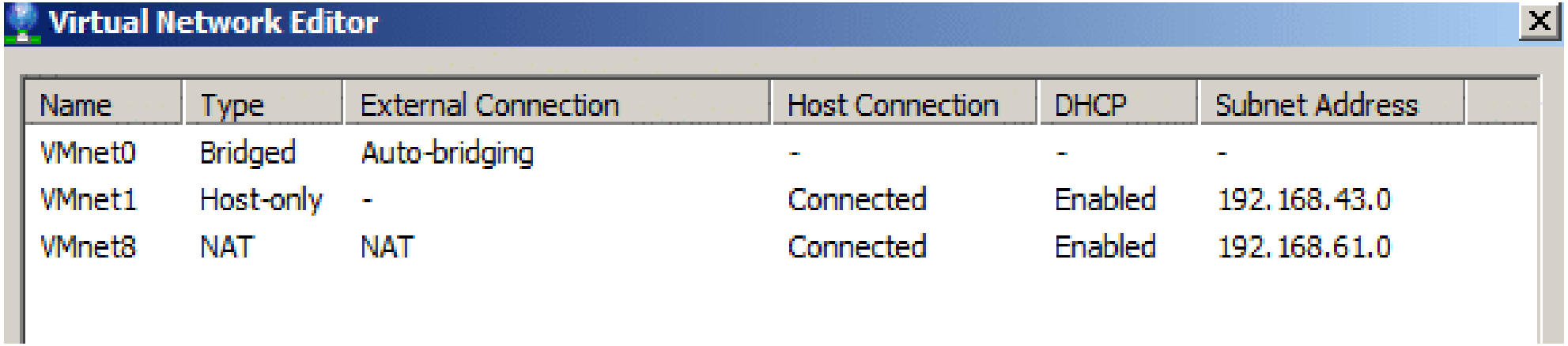
Network Modes in VMware Workstation

Bridge, NAT and Host-Only

Ameur Imane

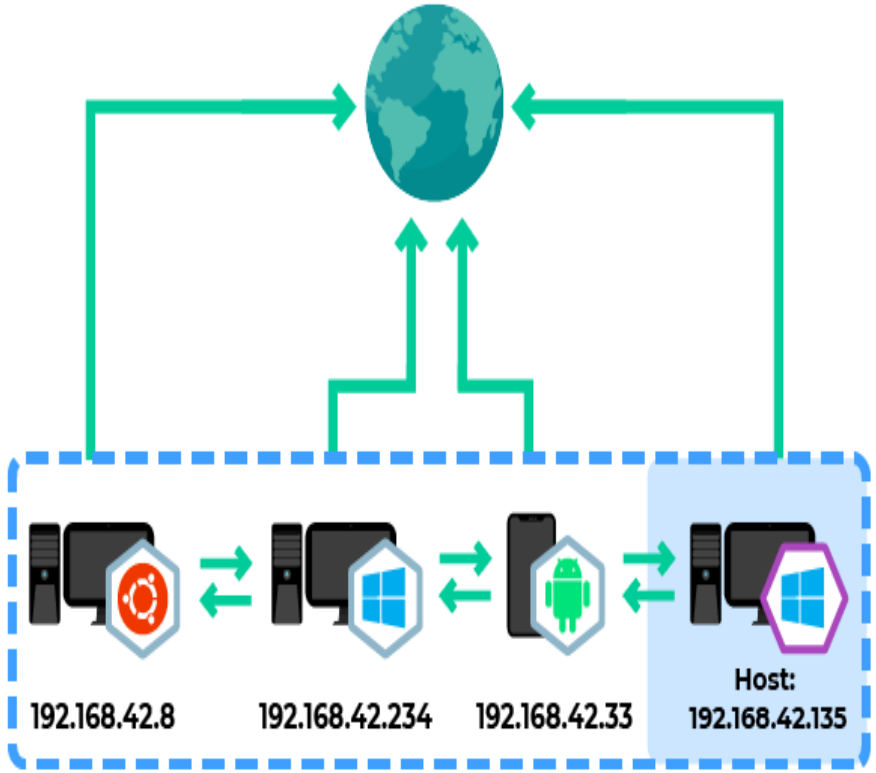
Introduction

- In VMware Workstation, several network configuration options are available to connect virtual machines in different ways, depending on communication and access needs. Here is an overview of the Bridged, NAT, and Host-Only modes.



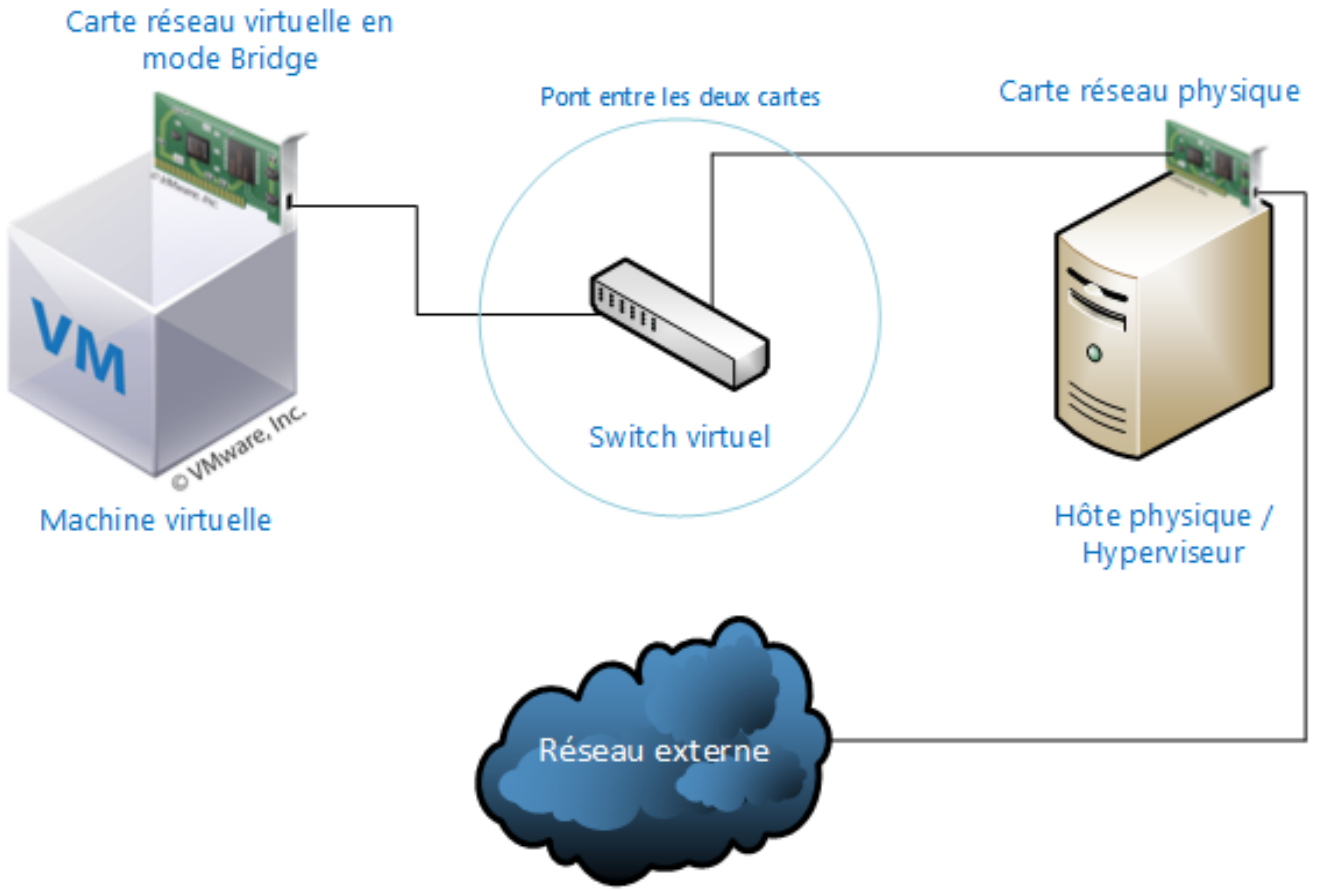
1 Bridged Networking (Accès par Pont)

- **What is it ?**
- In this mode, the virtual machine (VM) behaves exactly like a physical “host” computer connected to your local network. It obtains its own IP address on the same network as the host, usually via the DHCP server of your router.



- **How it works:**

The virtual network adapter connects directly to your physical network adapter. The VM will request its own IP address directly from your router (Internet box).



A **bridge** is created between the virtual network adapter of the virtualization application and the network adapter of your physical host.

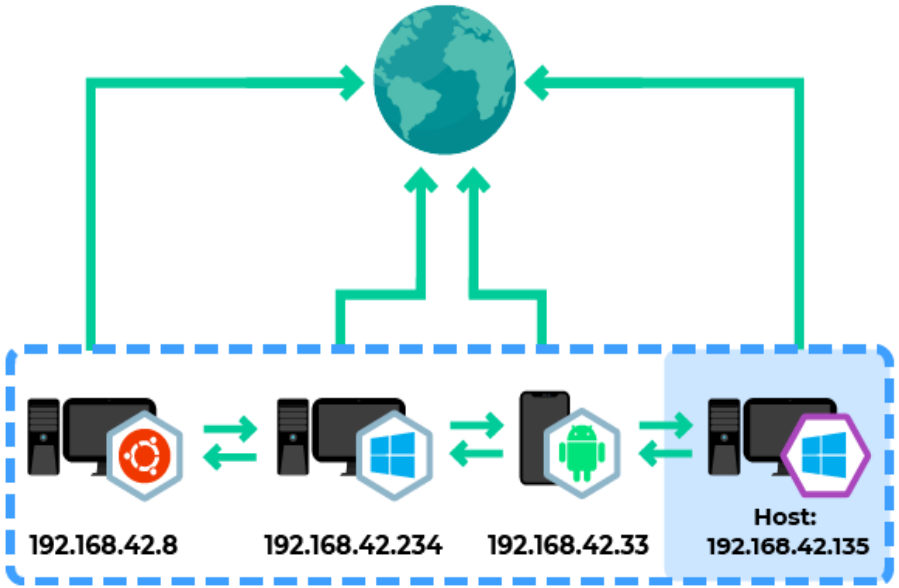
1 Bridged Networking (Accès par Pont)

- **Visibility:**

The VM is visible to all other machines on the network (your PC, your colleagues' PCs, your network printer, etc.).

- **Analogy:**

It's like plugging an extra Ethernet cable from the VM directly into your router. The VM is a "housemate" in your network.



1 Bridged Network Mode

Advantages

- Ideal for servers or services that need to be accessible on the LAN.
- Operates naturally, just like a physical machine.

Disadvantages

The VM is exposed to other devices on the local network, which can pose security risks (less isolation → higher security risk).

May cause issues on certain Wi-Fi connections (bridging can be limited).



1 Bridged Networking (Bridge Access)

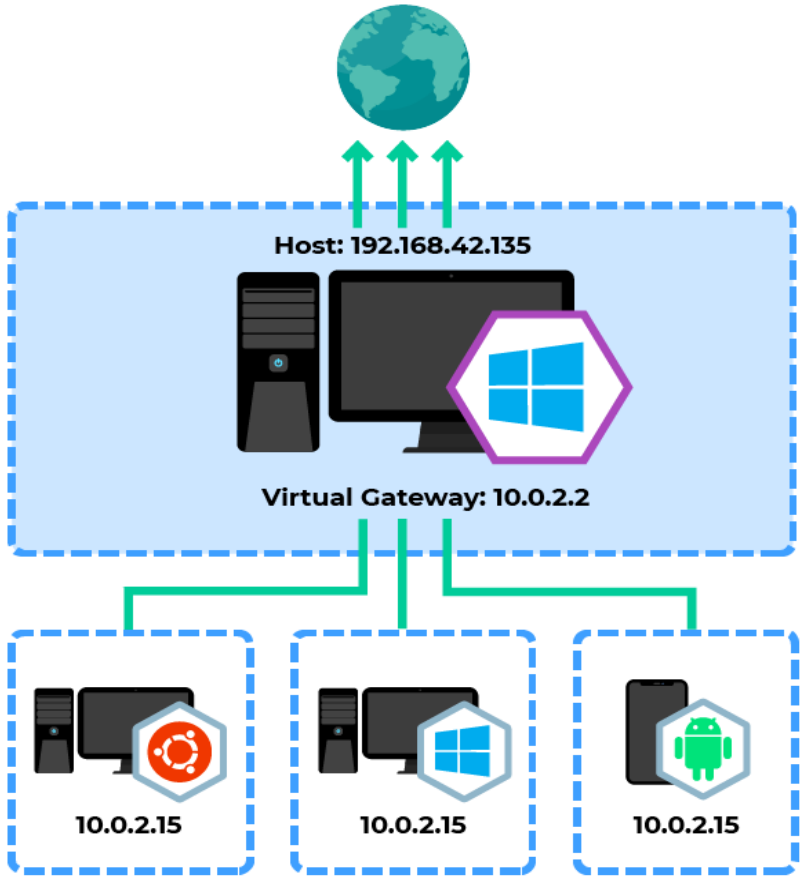
- **When to Use:**
- When you want the VM to be accessible by other computers on the network (e.g., a web server).
- When you need to easily share files between the VM and other PCs.

2 NAT (Network Address Translation)

What is it?

- This is the default mode in most virtualization software. The VM shares the host's IP address and connects to the external network through the host.

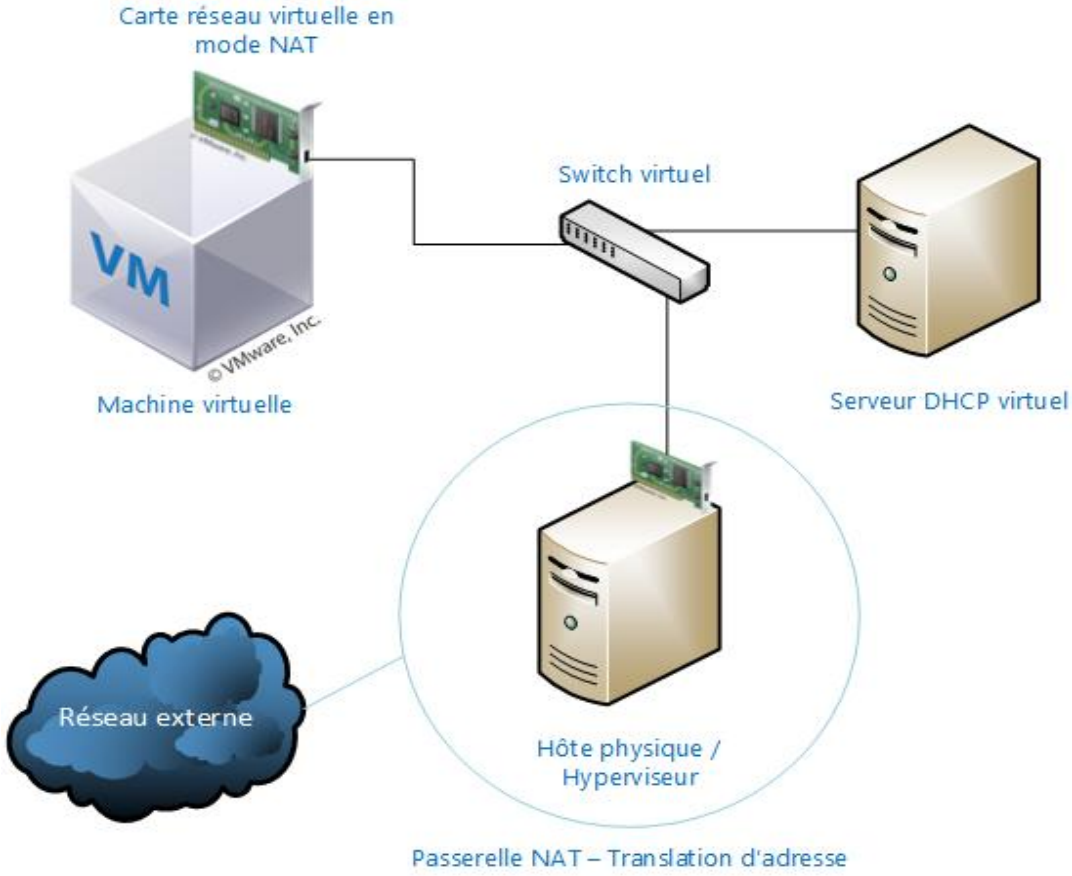
- From the perspective of the outside world, the traffic appears to come from the host machine.



2 NAT (Network Address Translation)

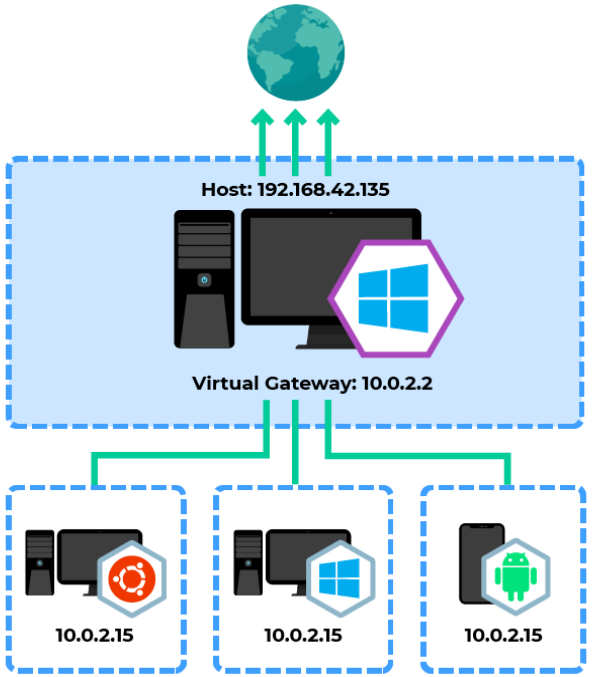
How it works:

The virtual machine creates a private subnet inside your host computer. To access the Internet, it “hides” behind your host’s IP address. The physical router does not even know that the VM exists.



2 NAT (Network Address Translation)

- **Visibility:**
  The VM can access the Internet and the network, **but the outside world cannot access the VM** (unless you set up port forwarding).
- **Analogy:**
  Your host computer acts as a router, and the VM is a device connected behind it. It’s like a **sub-tenant** using your address to send mail.



2 NAT (Network Address Translation)

Advantages

- Simple and secure.
- Ideal if the VM needs Internet access but should not be visible on the local network.

Disadvantages

- The VM is not visible from the local network, which limits interactions with other devices outside the host.



2 NAT (Network Address Translation)

When to Use:

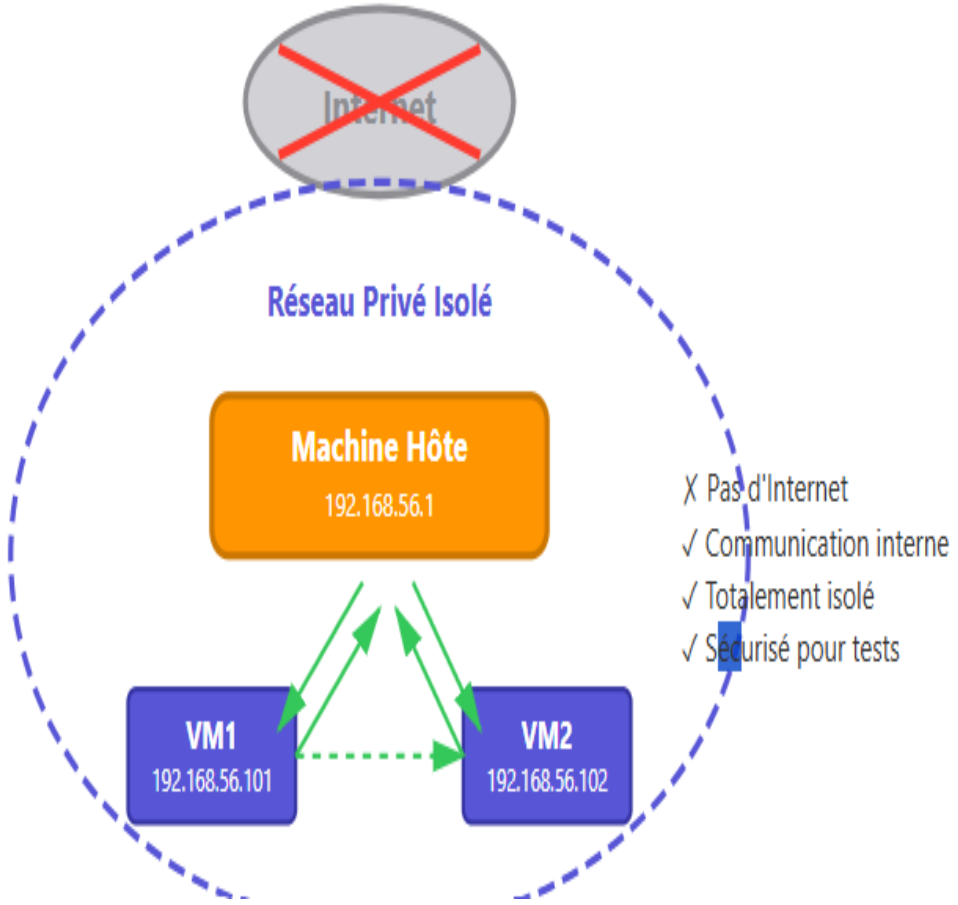
- When you only need Internet access on the VM (browsing, updates).

- When you want to protect the VM from attacks coming from the local network (basic isolation).

3 Host-Only (Host-Private Network)

What is it?

- This mode is the most restrictive and isolated.
- It creates a private network between the host and the VMs. **The VMs can communicate with the host and with each other**, but they cannot access external networks or the Internet (unless the host is configured to route the traffic).

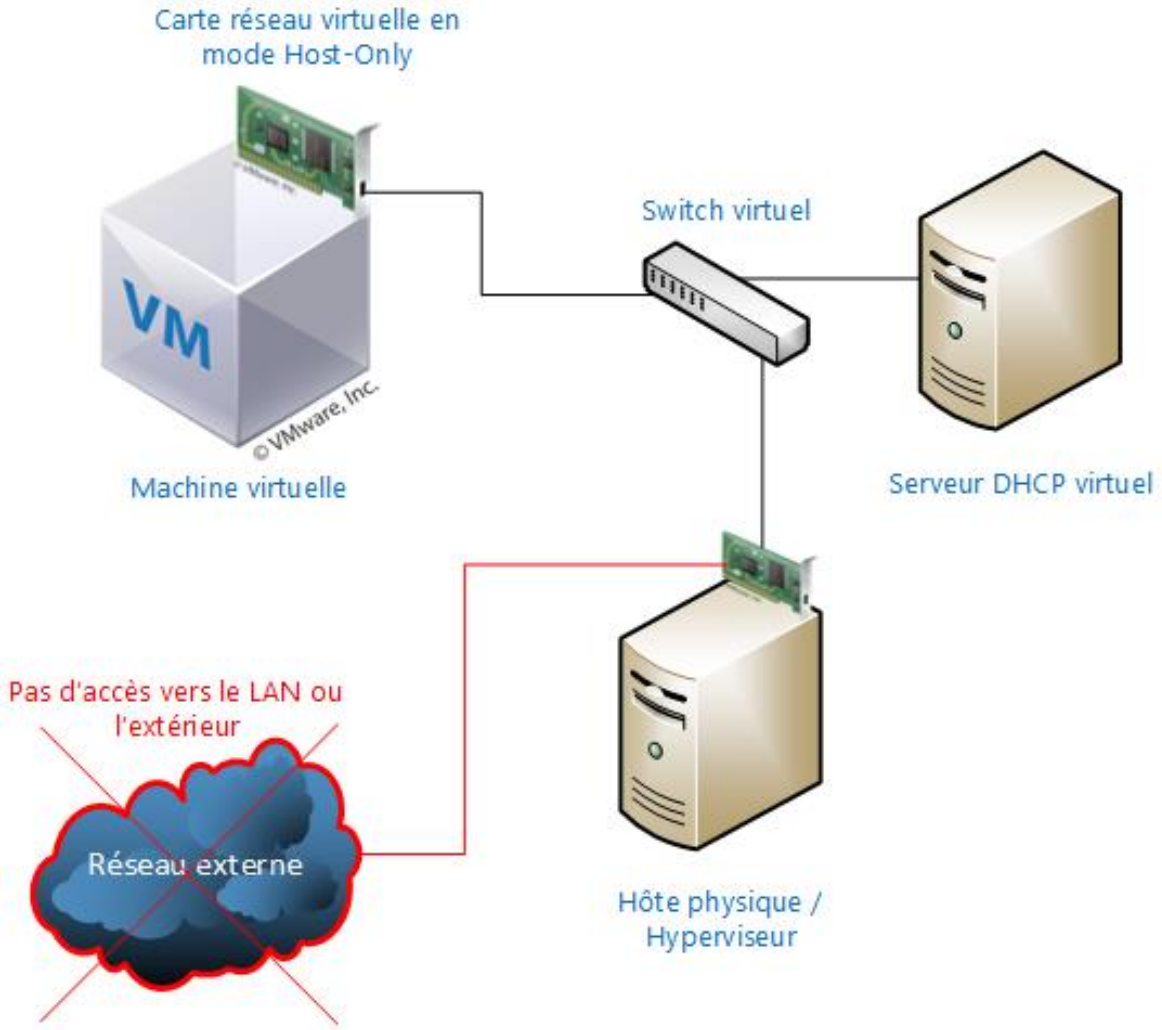


3 Host-Only (Host-Private Network)

How it works:

A network is created exclusively between your host computer and the VM.

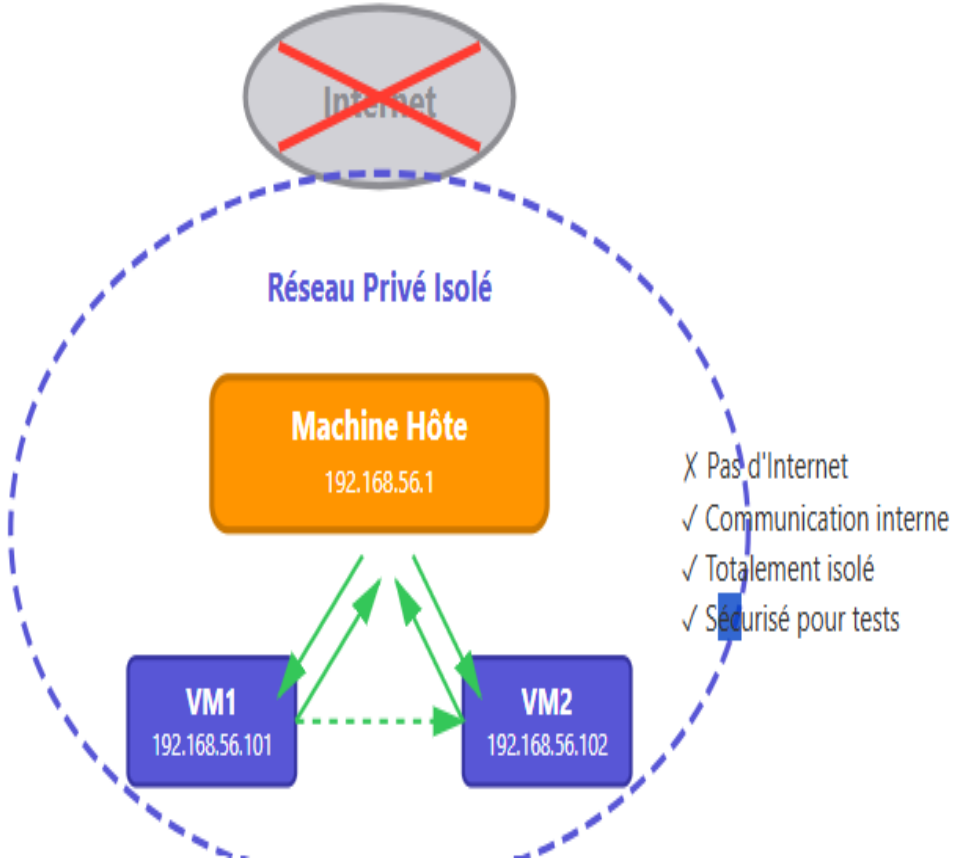
The physical router and the Internet are excluded from the setup.



3 Host-Only (Host-Private Network)

- **Visibility:**
  The VM can communicate with the host, and the host can communicate with the VM. No one else can access it, and the VM does not have Internet access (by default).

- **Analogy:**
  It’s like connecting two computers with a cable in a locked room, without any connection to the outside world.





3 Host-Only (Host-Private Network)

When to Use:

- When you want to create a secure testing environment (sandbox) to analyze viruses or malware without risking infecting the network.
- When you are setting up a complex internal network between multiple VMs without needing Internet access.

3 Host-Only (Host-Guest Private Network)

Advantages

- Highly secure.
- Perfect for isolated testing, and lab environments.

Disadvantages

- No Internet access by default.
- Mainly useful for specific scenarios.



Summary

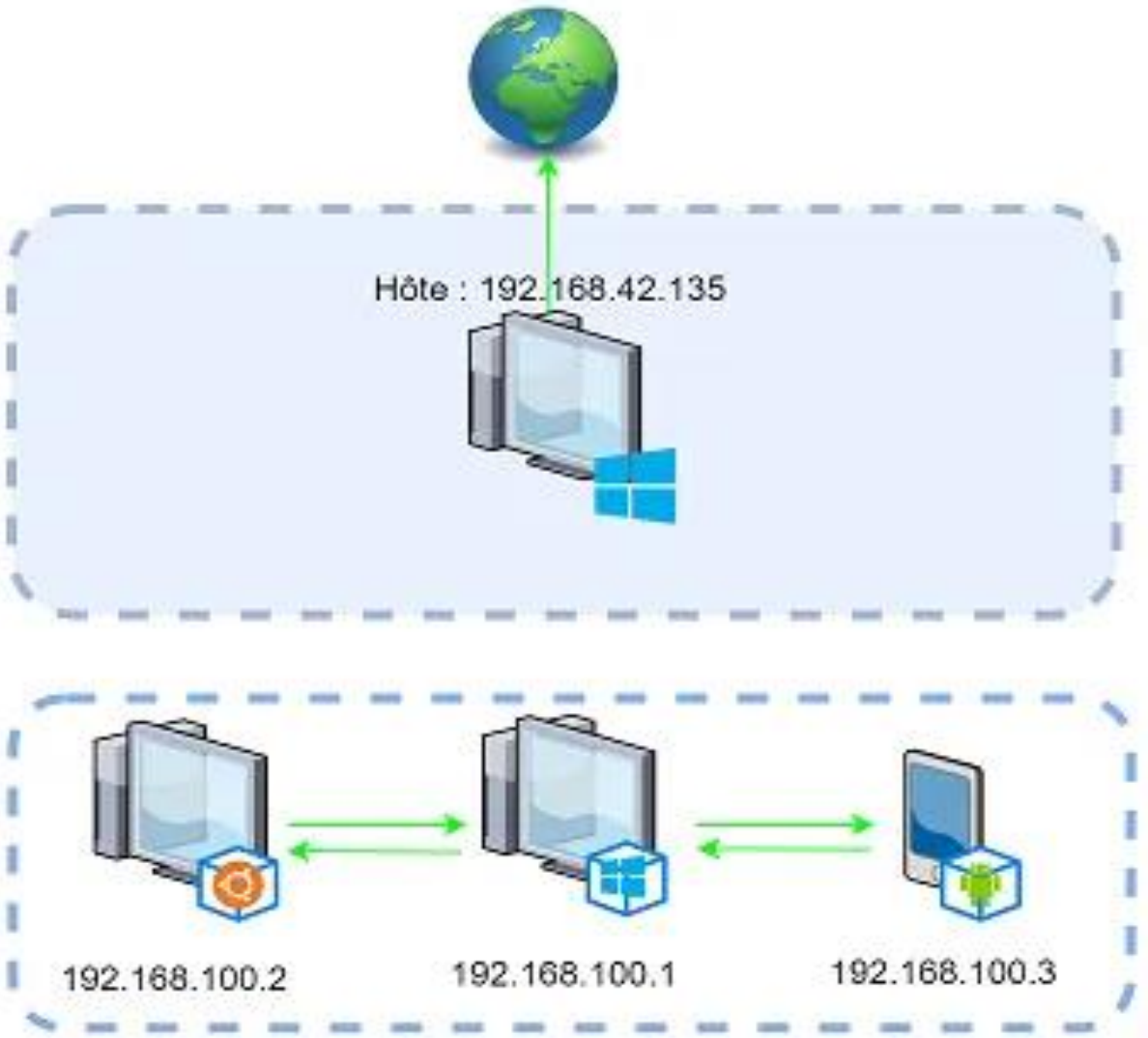
Mode	Internet	LAN Access	Host Access	Visibility
NAT	✔	✘	✔	VM not visible from the LAN
Bridged	✔	✔	✔	VM visible as a network PC
Host-Only	✘	✘	✔	VM visible only by host

Summary

Characteristic	NAT	Bridged	Host-Only
Internet Access	✅ Yes	✅ Yes	❌ No
Access from External Network	❌ No (except with port forwarding)	✅ Yes	❌ No
VM ↔ Host Communication	✅ Yes	✅ Yes	✅ Yes
IP Address Type	Private (10.x.x.x)	Local Network (192.168.x.x)	Private (192.168.56.x)
Ideal Use Case	Daily Use, Browsing	Servers, Network Services	Isolated Testing, Development
Security Level	⭐ High	⭐ Medium	⭐ Very High

LAN segment

Allows VMs to connect to eac**without going through a virtual switch**h other. Useful for isolating a group of VMs. This is the configuration we will use to simulate a complete IT infrastructure.



Internal Network

Similar to the **host-only** network, but without a connection to the host.
The VMs can communicate with each other, but they are completely isolated from the host and the outside world.
You must configure the IP addresses of your VMs manually using static IP settings.

